Pricing and Risk

Where every number comes from, and every case it has to survive

FRE-GY 6931 Blockchain & Cryptocurrency, Fall 2026 — Team 9
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How to read this document

There are two classes of number in this project, and keeping them apart is the whole point of this document.

The contract's parameters were chosen. Eight values are fixed in the deployment, and five of them set the economics: the 10bp vault discount, the 20%/10% initial and maintenance margin pair, and the 1%/3% collateral haircut. They were set by engineering judgement when the contracts were written, committed before any of the analysis below existed. Four of the five are immutable on Sepolia; the discount alone can be raised by the owner, up to a hard cap set at deployment. They are the *subject* of this analysis, not its output.

Everything in this document is derived. The 6.28bp requirement, the 32.9bp break point, the 0.456-year duration limit, the 3.98% typical closure and every figure in Section 9 come out of the two data series in §1.1. None was chosen, and none was adjusted to flatter a parameter.

That ordering — parameters first, measurement second — is the honest one and also the demanding one: the analysis could have found the chosen values indefensible, and had to be run in a way that would have shown it. This document does three things with them.

1. **Shows where each one came from** — which data series, which formula, which judgement call, and which of those judgements is doing the most work.
2. **Walks every case** — from a median weekend to the worst weekend in eleven years of data, in dollars, one step at a time.
3. **Says where each number breaks** — the exact input value at which it stops working, and whether history has already reached that value.

Section 9 is a line-by-line provenance table: every figure quoted anywhere in our write-up, its source, and the one line of arithmetic that produces it. Nothing in this document was typed by hand; Appendix A reproduces all of it from the cached data files.

Two results are worth stating up front, because they are what the exercise was for.

The fee is correctly sized, and we can prove where it stops being correct. 10bp against a 6.28bp requirement is 1.59× coverage, and we locate the break point of every input that feeds it. Two of the four matter; the other two have two to three times the headroom they need and can be argued about freely without changing a conclusion. Knowing which is which is the difference between a priced product and a guessed one, and it is also why one candidate term is deliberately absent: a risk we cannot estimate is named in §8.3 rather than given an invented number in the fee.

And the analysis found the binding constraint, which is not the fee. Across all seven historical crises we tested, what threatens the system is the vault running out of cash — a liquidity limit that sits in the vault's *sizing*, not in its pricing. That is a more valuable finding than a confirmation would have been: it moves the design question from "is 10bp enough?", which we can now answer yes, to "how much capital stands behind it?", which is where the remaining work belongs. Section 8 develops both.

1 Inputs

1.1 Data

Four series, cached in `docs/data/` so every figure is reproducible without a network call.

Series	Source	Coverage	Used for
3-month T-bill yield (DTB3)	FRED, daily, discount basis actual/360	2015-01-02 to 2026-10-23, 3,085 obs	Weekend yield shocks; the current yield y
BTC/USD hourly	Coinbase, OHLC	2021-01-01 to 2026-10-23, 51,070 bars	Closure moves, regime stress, liquidation tests
BTC perp funding, hourly	Deribit	matched to the above	Funding-rate realism check
Sepolia reconciliation	our own deployment	28 reads	Confirming the settings below match the deployed contracts

1.2 Settings

These are not assumptions. Each is read back off the deployed contract in Section F of the notebook and matched against the value used here.

Setting	Value	Where it binds
Initial margin	20%	Caps opening leverage at 5×
Maintenance margin	10%	Liquidation trigger
Haircut, market open	1%	sBILL counted at 99% of NAV
Haircut, market closed	3%	sBILL counted at 97% of NAV
sBILL interest	4.00% simple	Accrues on wall-clock, weekends included
Vault discount	10bp	Vault buys sBILL at NAV × 0.999. Owner-adjustable up to a 1% cap, only while the market is open
Liquidation reward	50bp	Paid to the liquidator
Funding cap	10bp/hour	Largest rate the owner can post

1.3 Judgement parameters

Six inputs are genuine judgement calls. Five of them are currently buried as function defaults in the notebook rather than declared alongside the settings — a presentation flaw we name here because these are precisely the numbers a reader should be allowed to disagree with.

Parameter	Value	Basis
Closure length	65 hours	Friday 4pm ET to Monday 9am ET. `MarketCalendar` closes Friday 21:00 to Monday 14:00 UTC, the same 65 hours, so contract and analysis agree.
Hurdle premium	2.00%	What a vault depositor should want *above* T-bills for taking contract and liquidity risk. Assumed, not observed.
Collateral duration	0.25 yr	Assumes 3-month paper. Section 4 shows this is the tightest assumption in the model.
P(long weekend)	9/52	Nine of fifty-two closures are extended by a Monday holiday or Good Friday.
Extra hours if long	24	One additional day of capital lock-up.

2 What actually happens in a closure

Before the arithmetic, the mechanism. A perpetual swap settles hourly. When the market is open and a trader's margin falls short, the clearinghouse redeems their sBILL with the issuer at full NAV. When the market is closed the issuer is unreachable, so the clearinghouse sells the sBILL to our vault instead, at a discount.

1. **Friday 4pm ET.** The T-bill market closes. The clearinghouse switches the haircut from 1% to 3%, so every sBILL-backed position is now counted slightly more conservatively.
2. **Each hour of the closure.** The price moves. The losing side owes cash and holds none, so `_collect()` routes to the vault: the vault pays cash, and takes sBILL worth slightly more than the cash it paid.
3. **Monday 9am ET.** The issuer reopens. The vault calls `redeemInventory()`, hands the sBILL back at NAV, and books the difference.

The vault is therefore a weekend *liquidity* provider, not a trader. It earns the discount, earns the T-bill interest for the hours it holds the paper, and carries the price risk of that paper across the closure. Those three terms are the whole pricing problem.

2.1 The decomposition

Per $1 of cash the vault actually deploys, over a closure of h hours:

$$\underbrace{\frac{(y+\pi)h}{8760}}_{\text{gross carry}} - \underbrace{\frac{yh}{8760}}_{\text{sBILL accrual}} + \underbrace{D \cdot \Delta y}_{\text{price risk}} + \underbrace{p_\ell \frac{(y+\pi)h_{\text{extra}}}{8760}}_{\text{long-weekend risk}} = \text{required discount}$$

with y the T-bill yield, π the hurdle premium, D duration, Δy the weekend yield shock, and the last term the cost of a Monday holiday extending the closure.

Δy is an **empirical percentile, not a standard deviation**, and the choice is deliberate. The standard deviation of weekend moves is 4.17bp, but the series is far from normal: a normal distribution expects 0.04 weekends beyond 4σ in a sample this size and we observe nine, and June 2022 is an 8.4σ move that a normal distribution says cannot occur. Sigma-scaling would therefore understate the tail by orders of magnitude. We read the quantile straight off 597 observed weekends instead, which assumes nothing about the shape of the distribution. The base case uses the 99th percentile, 18bp — 6.5 times the average absolute move and 4.3 times one standard deviation.

2.2 What is being priced

One convention governs every term above, and it is worth stating before the numbers rather than after them: **we price the collateral the system is designed for, not the mock standing in for it.**

Our sBILL accrues a fixed 4% on a clock, so rates cannot move its value. The dominant term $D \cdot \Delta y$ therefore prices a risk our own deployment cannot experience, and that is deliberate: the fee has to be right for the asset the design calls for, not for the stand-in. Dropping it on the grounds that the mock cannot produce it would leave $1.48 + 0.28 = 1.76$bp against the 10bp charged, and would price a demonstration rather than a design.

The same convention has a limit, and we draw it at measurability. A risk enters the fee only when it can be estimated from something. The weekend yield shock is read off 597 observed weekends; the long-weekend term is counted off a calendar, nine closures in fifty-two. **Redemption gates meet neither test.** Issuer discretion over redemption is a real feature of tokenized Treasury products, and our own `SBill` has a structural analogue — it pays interest from a reserve funded once at deployment, and `redeem()` transfers without checking the balance, so an

exhausted reserve reverts the redemption. But the sector is roughly two years old and has never been stressed, so any frequency we assigned would be a prior wearing the costume of an estimate. We therefore **name this risk and do not price it**, in §8.3, alongside the larger unpriced exposure in §8.4. Carrying it in the fee would have added 0.29bp and changed no conclusion in this document.

Note the first two terms partly cancel: the vault's capital must earn $y + \pi$, but the sBILL it holds is itself earning y, so only the premium π has to come out of the discount. This is why the net carry term is small.

2.3 Weekend yield shocks

Measured from each Friday's published DTB3 rate to the next published rate, keeping only gaps of four days or fewer — 597 weekends.

	Yield move	**Price impact at $D = 0.25$**
Median	2bp	0.50bp
95th percentile	10bp	2.50bp
99th percentile	18.04bp	4.51bp
Largest, 2022-06-10	35bp	8.75bp

The four largest weekend moves are 2022-06-10 (+35bp), 2023-03-17 (+29bp), 2022-07-08 (+22bp) and 2022-10-07 (+22bp). All four fall in the 2022–23 tightening cycle, which is the regime this pricing should be judged against.

2.4 The three cases

With $y = 4.01\%$ (DTB3, 2026-10-28) and $h = 65$:

Case	**Hurdle**	**Duration**	**Shock**	**Terms (bp)**	**Required**
Low	4.01%	1 month	10bp	$0.00 + 0.83 + 0.19$	1.02bp
Base	6.01%	3 months	18bp	$1.48 + 4.51 + 0.28$	**6.28bp**
Stress	9.01%	3 months	35bp	$3.71 + 8.75 + 0.43$	12.89bp

The contract charges **10bp**: 1.59× the base requirement, and comfortably clear of the low case. The stress case at 12.89bp sits above it, which is the useful result rather than an awkward one — it means the fee has a locatable limit, and Section 4 pins that limit to the exact value of each input. A fee that covered every conceivable case would simply be overpriced.

Observe which term dominates: T-bill *price risk* is 4.51 of the 6.28bp, 72% of the total. The single largest component of our fee prices the risk that interest rates move while the vault is holding the paper.

3 One trade, five cases

Take $100,000 of collateral routed through the vault in a single closure. Fee and carry are fixed; only the rate shock varies.

Weekend	**Shock**	**Fee**	**Carry**	**Price loss**	**Kept**	**vs hurdle $47.44**
Typical (median)	2.0bp	$100.00	$29.75	−$5.00	$124.75	+$77.31
Bad, 1-in-20	10.0bp	$100.00	$29.75	−$25.00	$104.75	+$57.31
Bad, 1-in-100	18.04bp	$100.00	$29.75	−$45.10	$84.65	+$37.21
Break point	**32.9bp**	$100.00	$29.75	−$82.31	$47.44	**$0.00**
Worst ever (Jun 2022)	35.0bp	$100.00	$29.75	−$87.50	$42.25	−$5.19

Each row reads the same way. The vault pays out $100,000 in cash and takes back sBILL worth $100,100, so the fee is $100. It holds that paper for 65 hours at 4.01%, which is $29.75. Then rates move, the paper loses $D \times \Delta y$ of its value, and whatever remains is what the vault keeps. Its depositors required $47.44 for that weekend.

3.1 What "breaks" means

The word is doing a lot of work, so it is worth separating two thresholds that are easy to collapse into one.

	Missing the hurdle	**Losing money**
What happens	Vault profits, but by less than depositors required	Vault ends the closure with less than it started
Threshold, 3-month paper	32.9bp	51.9bp
Reached in history?	Yes — once, June 2022 at 35bp	No. Largest ever is 35bp
Who notices	Depositors earn below hurdle	ERC-4626 share price falls

So the correct statement about June 2022 is narrow: on $100,000 the vault came up **$5.19 short** of what it had promised, having still made $42.25. It is a knife-edge, not a cliff. Nobody lost principal.

4 Where each input breaks

Each row solves one input for the value at which 10bp stops clearing the hurdle, holding the others at base.

Input	**Base**	**Breaks above**	**Headroom**	**Reached in history?**
Weekend yield shock	18bp	**32.9bp**	1.83×	**Yes — 35bp, Jun 2022**
Collateral duration	0.25 yr	**0.456 yr**	1.82×	Immediately, on 6-month paper
Closure length	65 h	228 h	2.0× vs 113 h	No. Longest closure is 113 h
Hurdle premium	2.0%	6.7%	3.4×	Not observable

Two inputs matter and two do not. The closure length and the hurdle premium have two to three times the headroom they need, so arguing about them changes nothing. The yield shock has already been breached once, and the duration assumption has the tightest margin in the model — and unlike the other three it is an assumption about our own collateral rather than about the world, which makes it the one we can actually do something about.

4.1 The duration assumption

This is the one carrying the most weight, because it is an assumption about the collateral rather than about the world. Price sensitivity is linear in duration, so doubling the maturity doubles the loss from any rate move.

Collateral	**Loses above**	**Worst ever**	**Verdict**
3-month bill (our assumption)	51.9bp	35bp	1.48× headroom
6-month bill	26.0bp	35bp	**Has happened**
1-year note	13.0bp	35bp	**Happens regularly**; 99th pct is already 18bp

The design implication is concrete and belongs in the contract rather than the spreadsheet: **the vault should accept only short-dated collateral, and that limit should be enforced, not assumed.** At 3 months the pricing survives the worst weekend on record. At 6 months it does not survive an ordinary bad one.

5 Case study: June 2022, the worst weekend

This is the single most informative closure in the data, because both legs of the risk fired at once.

On Friday 10 June 2022 the May CPI print landed at 8.6%, the highest since 1981. Rates repriced through the close:

Thursday 9 June	1.26%
Friday 10 June	1.34%
Monday 13 June	**1.69%**
Move across the closure	**+35bp**

Meanwhile BTC fell from $29,169 at Friday 4pm ET to a low of $23,319 and closed Monday at $23,610 — a 19.06% adverse excursion on settled hourly prices, with no recovery inside the closure.

That combination is what makes it the worst case rather than merely a bad one: **the vault buys the most collateral at exactly the moment that collateral is worth the least.** The two risks are positively correlated, which is the opposite of what a liquidity provider wants.

On $1m of notional, a long would have sold roughly $190,567 to the vault:

Collateral	Fee	Carry	Price loss	Net	Dollars
3-month bill	+10.00bp	+0.99bp	−8.75bp	+2.24bp	+$42.77
6-month bill	+10.00bp	+0.99bp	−17.50bp	−6.51bp	−$123.98
1-year note	+10.00bp	+0.99bp	−35.00bp	−24.01bp	−$457.47

Note the carry: only 0.99bp, against 2.98bp at today's yields. Rates were 1.34% in 2022, so the cushion that normally absorbs a shock was mostly absent. **The pricing is weakest when rates are low and moving** — which is also when a 35bp weekend becomes possible. A third correlation working against us.

6 Regime stress

For each crisis we take the worst closure inside the surrounding window, measured as the adverse excursion from Friday 4pm ET over 65 hours on settled hourly closes. Excursion is the right measure rather than the Friday-to-Monday net move, because settlement is hourly: the losing side pays as the price goes against it, and a later recovery does not return the collateral already sold.

Regime	Weekend	BTC open	Excursion	Sold/$1m	Fee	×typ
Yen carry unwind	2024-08-02	$62,553	20.38%	$203,816	$204	5.1×
Jun 2022 CPI shock	2022-06-10	$29,169	19.06%	$190,567	$191	4.8×
SVB / USDC depeg	2023-03-10	$19,972	13.54%	$135,378	$135	3.4×
May 2021 mining ban	2021-05-14	$49,193	12.80%	$127,973	$128	3.2×
LUNA/UST collapse	2022-05-06	$36,081	9.10%	$91,048	$91	2.3×
Oct 2025 deleveraging	2025-10-24	$110,920	4.63%	$46,333	$46	1.2×
FTX collapse	2022-11-11	$16,573	4.45%	$44,540	$45	1.1×
Typical weekend	—	—	3.98%	$39,795	$40	1.0×

This table measures excursions on hourly closes from Friday 20:00 UTC. The report's Section 4 table, written by `docs/stress_weekends.py`, uses the notebook's convention (hourly opens, Friday 4pm New York time) and takes the worst closure within three weeks of each event, which selects 10 October 2025 rather than the 24th. Its figures differ from these by small amounts (for example 18.33% rather than 19.06% for June 2022, and a 95th-percentile capacity of $98,461 rather than $93,522), and the conclusions below hold under both.

6.1 Reading the revenue column honestly

Crises are good for the vault's revenue line and bad for its risk. Volume scales with the move, so the yen unwind would have earned 5.1× a typical weekend. But the fee is 10bp of a larger number — $204 instead of $40 per $1m of notional — and nothing about that $204 compensates for the two problems the same weekend creates.

First, capacity. Sizing the vault at the 95th percentile of weekend excursions gives $93,522 of capacity per $1m of notional. Four of the seven crises exceed it:

Regime	Needs	% of capacity
Yen carry unwind	$203,816	218%
Jun 2022 CPI shock	$190,567	204%
SVB / USDC depeg	$135,378	145%
May 2021 mining ban	$127,973	137%
LUNA/UST collapse	$91,048	97%
Oct 2025	$46,333	50%
FTX collapse	$44,540	48%

A 95th-percentile vault is undersized for exactly the weekends it exists to handle. When it empties, settlement stops and waits for Monday.

Second, liquidation. Whether the position survives depends entirely on opening leverage, and the 3% haircut barely moves the threshold.

Leverage	Long liquidated below	Short liquidated above	Crises surviving
2.6× (demo)	−31.4%	+25.5%	all seven
4.0×	−16.4%	+13.3%	four of seven
4.85× (max)	−11.5%	+9.3%	three of seven

At the demo's 2.6× nothing is liquidated in any regime. At 4× the June 2022 and yen-unwind longs go, and the SVB short goes. At the maximum the 2021 mining ban takes the long too. The demo is safe because it is conservatively levered, not because the parameters are strong — a distinction worth stating plainly rather than letting the Sepolia run imply otherwise.

7 The haircut

A haircut counts collateral at less than face value, because it is not cash and would have to be sold. Ours is 1% when the market is open and 3% when it is closed. The logic is right: on a weekday you redeem with the issuer at full NAV, while at a closure your only buyer is the vault, at a discount — or there is no buyer at all.

7.1 What it costs

$30,000 posted as	**Counts as**	**Max leverage**	**Short liquidated at**
Cash (tUSDC)	$30,000	5.00×	$98,049 (+25.94%)
sBILL, 1% haircut	$29,700	4.95×	$97,959 (+25.82%)
sBILL, 3% haircut	$29,100	4.85×	$97,774 (+25.59%)

Moving from the weekday to the weekend haircut brings the liquidation threshold **$185 closer** on a figure near $98,000 — 0.19% of it — and costs a tenth of a turn of leverage. The weekend haircut is close to free.

7.2 What it is sized against

Measured T-bill price risk, 99th pct	4.51bp
Worst-ever price risk, 35bp shock	8.75bp
Vault discount paid on conversion	10.00bp
Total cost of converting, worst case	18.75bp
The haircut	**300bp**
Multiple of measured price risk	67×
Multiple of worst-case conversion cost	16×

So 3% is one to two orders of magnitude larger than the valuation risk it is usually described as covering — and measuring that gap is what tells us the haircut is doing a different job from the one its name suggests.

Keep it, and describe it correctly. It costs $185 and a tenth of a turn of leverage, and it is the one place in the contract that becomes more conservative the moment the market shuts, which is exactly the right instinct. What it buys is not a calibrated price for 4.5bp of measurable T-bill price risk; it is deliberately generous cover for the closure risks we cannot put a number on — a longer closure, a delayed redemption, an issuer problem. At $185 that cover is close to free, which is the strongest possible argument for keeping it.

7.3 What it cannot cover

The haircut answers "will this sell below face value?" — and over a weekend the answer is yes, by about 10bp. Trivial. The question that matters is "will it sell at all?"

BTC move over a 65-hour closure	**Size**	**vs 3% haircut**
99th percentile, short side	13.50%	4.5× uncovered
99th percentile, long side	13.93%	4.6× uncovered
Worst observed	20.38%	6.8× uncovered

This is not really the haircut's failure. The contract has two distinct buffers, and they answer different questions:

- **The haircut (3%)** — this asset may not convert at face value. A *valuation* adjustment.
- **Maintenance margin (10%)** — the price may move before the position can be closed. A *solvency* buffer.

A vault with no cash left is a *liquidity* failure, and neither buffer prices it. Marking the collateral down does not produce a buyer. That risk is carried by vault sizing, and Section 6 shows the sizing is the weak link.

8 What this pricing does not cover

The analysis establishes three boundaries on its own results. Each one is a finding in its own right, and each points at a specific change rather than a caveat.

8.1 The model is deliberately more conservative than the implementation

`SBill.nav()` grows purely on the clock:

```
1e18 + (1e18 * apyBps * (block.timestamp - startTime)) / (10_000 * 365 days)
```

A fixed 4%, which never reprices when rates move. The 35bp spike of June 2022 would be invisible to our deployment: the vault would bank its full 10bp and notice nothing.

We priced the risk the collateral would carry in production, not the lighter risk the mock carries — which is the right way round, and worth being explicit about. T-bill price risk is 4.51 of the 6.28bp required discount, 72% of the total, and it is a risk our deployment *structurally cannot experience*. So the largest component of the fee is one that no unit test and no Sepolia transaction can exercise. The pricing is built for a real tokenized T-bill and will hold when one is swapped in; the demo is simply a weaker test than the model it implements.

8.2 A loss would be invisible on-chain

If the vault did take a price loss, it surfaces at redemption. The stored `inventoryCost` is zeroed, `totalAssets()` falls, and the ERC-4626 share price drops. Depositors absorb it. Nothing reverts and nothing pauses.

And the event cannot distinguish a loss from breaking even (`WeekendVault.sol:130`):

```
emit InventoryRedeemed(shares, usdcReceived, usdcReceived > cost ?  usdcReceived - cost
                                          :  0);
```

On a loss the profit field emits zero. A depositor reading the log cannot tell "broke even" from "lost $457." Emitting a signed `int256`, or emitting `cost` alongside `usdcReceived`, makes losses visible. A one-line fix, and finding a silent-loss path in one's own code is exactly what a security review is for.

8.3 Redemption gates: named, not priced

A tokenized fund can suspend redemptions. If that happened on a Monday the vault would hold sBILL it could not convert, for an unknown period — and our own `SBill` has the same shape of failure, since `redeem()` transfers from a reserve funded once at deployment and reverts when that reserve is empty.

We do not price it, and the reason is the measurability test in §2.2. Every other input in the fee is anchored to something countable: 597 observed weekends for the yield shock, fifty-two calendar closures for the long-weekend term. Tokenized Treasury funds are roughly two years old and have not been through a stress event, so there is no base rate to estimate from. Any

frequency we assigned would be a prior presented as a measurement, and that is a worse failure than leaving the term out.

Leaving it out is cheap. On the priors we had originally used — a gate once in twenty years costing 3% — the term contributes 0.29bp, about 4% of the fee, and removing it changes no conclusion in this document: coverage improves from 1.52× to 1.59× and every break point moves further away. A term that small, resting on a number that soft, does not belong in a price.

What replaces it is a threshold. The fee stops clearing its hurdle if gates reach roughly one weekend in eighty, or about once every eighteen months. That is the claim we are willing to defend — not that gates are rare at some particular rate, but that they are rarer than once every eighteen months, which the existence of a $15bn sector makes difficult to dispute. The exposure is managed by diversifying issuers and by sizing the vault, not by charging for it.

8.4 The inversion

The largest unpriced exposure is the one the vault holds all the time.

Priced: T-bill price risk	4.51bp
Priced: net carry	1.48bp
Priced: long-weekend risk	0.28bp
Total risk budget	**6.28bp**
Not priced: redemption gate (§8.3)	no base rate
Not priced: USDC depeg, March 2023	**1,300bp**
as a multiple of the entire budget	207×
as years of fee revenue, $1m book	89.5 years

The model prices four and a half basis points of risk on paper the vault holds for 65 hours, and zero on the stablecoin it holds permanently. In March 2023 USDC traded at $0.87. That is 1,300bp — more than two hundred times the whole risk budget.

It is also wrong-way. The vault holds the most cash during a crisis, because that is when settlement draws on it hardest — and a crisis is when stablecoins depeg. The SVB weekend appears twice in this document for that reason: as the third-largest draw in Section 6, and as the largest unpriced loss here. The same weekend maximises both. Circle's own stated trigger was that its wires had not settled before the weekend, which is the identical mechanism our system is built to address.

8.5 What we would change

1. **Cap collateral duration in the contract.** The tightest assumption in the model, and currently unenforced.
2. **Cap or reserve against the stablecoin leg.** We do not claim this exposure can be priced into the discount — a 1,300bp tail against a 6.28bp budget is not a charge any fee absorbs. It is a position limit: cap the vault's idle balance, hold part of it in a second stablecoin, or accept the exposure explicitly rather than silently.
3. **Size the vault above the 95th percentile.** Four of seven historical crises exhaust it.
4. **Emit signed profit and loss** from `redeemInventory()`.
5. **Promote the five hidden judgement parameters** into the settings cell, so a reader can disagree with them.

9 Figure provenance

Every number quoted in our write-up, with its source and the arithmetic that produces it.

Figure	Value	Source	How computed
Current 3m yield	4.01%	FRED DTB3, 2026-10-28	Last non-null observation
Weekend shock, median	2bp	FRED DTB3	Median $\lvert r_{t+1} - r_t \rvert$ over 597 Fri-to-next pairs
Weekend shock, 99th	18bp	FRED DTB3	99th pct of the same
Weekend shock, max	35bp	FRED DTB3, 2022-06-10	Max of the same
Hurdle per closure	4.74bp	Derived	$(y+\pi)\frac{65}{8760} + p_\ell(y+\pi)\frac{24}{8760}$
Required discount, base	6.28bp	Derived	$1.48 + 4.51 + 0.28$
Charged discount	10bp	`WeekendVault`	`discountBps`, read on Sepolia
Coverage ratio	1.59×	Derived	10/6.28
Carry on \$100k, 65h	\$29.75	Derived	$100{,}000 \times 0.0401 \times 65/8760$
Break point, yield shock	32.9bp	Derived	Solve $10\,\text{bp} + \text{carry} - 0.25\Delta y = 4.74\,\text{bp}$
Break point, duration	0.456 yr	Derived	Solve the same for D at $\Delta y = 18$bp
Principal loss, 3-month	51.9bp	Derived	Solve net kept $= 0$ at $D = 0.25$
Principal loss, 6-month	26.0bp	Derived	Same at $D = 0.5$
Jun 2022 BTC excursion	19.06%	Coinbase hourly	$1 - \min(\text{close})/\text{close}_{\text{Fri 20:00}}$, 65h
Jun 2022 net, 3-month	+2.24bp	Derived	$10 + 0.99 - 8.75$, carry at $y = 1.34\%$
Typical weekend	3.98%	Coinbase hourly	Mean worst-side excursion, 304 weekends
Worst weekend	20.38%	Coinbase, 2024-08-02	Max of the same
99th pct, long side	13.93%	Coinbase hourly	99th pct of down-excursion
Vault capacity, 95th pct	\$93,522	Coinbase hourly	95th pct excursion × \$1m
Haircut, counts as	\$29,100	Derived	$30{,}000 \times (1 - 0.03)$
Liquidation, weekend	\$97,774	Derived	$h(D+P_0)/(m+h)$, $h = 0.97$, $m = 0.10$
Haircut cost	\$185	Derived	$97{,}959 - 97{,}774$
Haircut vs price risk	67×	Derived	300/4.51
USDC depeg	1,300bp	Mar 2023, \$0.87 low	$(1 - 0.87) \times 10{,}000$
Depeg vs budget	207×	Derived	1300/6.28

A Reproducing these figures

All four data series are cached in `docs/data/`, so no network access is needed.

1. `docs/analysis.ipynb` Sections A–E produce the pricing decomposition, sensitivity grids and closure statistics. Section F reads the deployed Sepolia contracts and reconciles the settings in §1.2 against them; it is the only section that touches the network.
2. The settings in §1.2 are mirrored from the deploy script `00_deploy_weekend_margin.ts` in `packages/hardhat/deploy/`. They are not re-typed here: Section F asserts equality and fails loudly otherwise.
3. Figures in this document were recomputed independently of the notebook, from the same cached CSVs, as a cross-check. Two discrepancies surfaced and were resolved in favour of the stricter definition: closure moves are measured as *adverse excursion* rather than Friday-to-Monday net move, because hourly settlement draws on the vault at the worst point of the

closure and a later recovery does not return collateral already sold; and regime weekends are selected as the worst closure within a dated crisis window rather than hand-picked, so the choice is reproducible.

B Scope of the analysis

- **sBILL is a mock.** It accrues a fixed 4% from a pre-funded reserve. It demonstrates the integration; it does not carry real T-bill exposure, and as §8 sets out it cannot express the price risk that dominates the fee.
- **Day-count mismatch.** DTB3 is quoted on a discount basis, actual/360. sBILL accrues simple interest on 365 days. The difference is immaterial at 65-hour horizons but is a real inconsistency.
- **BTC history starts in 2021.** 304 weekends. The yield series reaches back to 2015 and 597 weekends, so the two samples are not co-extensive; the regime table is bounded by the shorter one.
- **Hourly closes, not trades.** Excursions are measured on settled hourly closes, matching how the contract settles. Intra-hour highs and lows are worse — the 99th percentile long excursion rises from 13.93% to 17.42% — so a liquidation engine reading a live price would fire more often than this analysis implies.
- **No holiday calendar.** `MarketCalendar` tests time-of-week only. On a trading holiday the contract believes the market is open, routes to a closed issuer, locks the vault out, and applies the gentle 1% haircut. With real collateral `redeem()` would revert and take `settle()` with it. The 113-hour Thanksgiving closure is priced in §4 but not implemented.
- **The hurdle premium is assumed.** 2% above T-bills is a judgement with no observable market price. It has 3.4× headroom, so it is not load-bearing.
- **Redemption gates are not priced.** The sector has no base rate, so the fee excludes them and §8.3 states the threshold at which they would matter instead. A reader who believes gates are more frequent than one weekend in eighty should treat the fee as uncovered for that risk.